

Cut-Expansion in Proof-Nets of Multiplicative Linear Logic

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Introduction

Cut-reduction

$$\frac{\frac{\pi_1}{\vdash B^\perp, A^\perp, \Gamma} \text{ (}\wp\text{)} \quad \frac{\frac{\pi_2}{\vdash A, \Delta} \quad \frac{\pi_3}{\vdash B, \Sigma}}{\vdash A \otimes B, \Delta, \Sigma} \text{ (}\otimes\text{)}}{\vdash \Gamma, \Delta, \Sigma} \text{ (cut)} \longrightarrow \frac{\frac{\pi_1}{\vdash B^\perp, A^\perp, \Gamma} \quad \frac{\pi_3}{\vdash B, \Sigma}}{\vdash A^\perp, \Gamma, \Sigma} \text{ (cut)} \quad \frac{\pi_2}{\vdash A, \Delta} \text{ (cut)}{\vdash \Gamma, \Delta, \Sigma}$$

Very useful

Example of result: can reduce to a cut-free proof (weak normalization)

Applications:

- No proof of falsity $\perp$ or 0
- Proof search
- ...

Introduction

Cut-expansion

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Useful?

Example of result: can expand to a proof with only one cut-rule, at its root

Applications:

- Craig's interpolation – following [Sau25]¹ and [FOS25]²
- Denotational semantic – used in [EFP24]³

¹ Interpolation as Cut-Introduction: On the Computational Content of Craig-Lyndon Interpolation

² On Correctness, Sequentialization and Interpolation

³ Bayesian Networks and Proof-Nets: a proof-theoretical account of Bayesian Inference

Introduction

Cut-expansion

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This talk: Simple proof of this result in *proof-nets*, adaptable enough to be transferred in most fragments of LL

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- ▶ Proof-Nets & Cut-Reduction
- ▶ Expanding to a unique *cut* & Applications
 - Craig's Interpolation
 - Denotational Semantic
- ▶ Proof of the Expansion to a unique *cut*
- ▶ Extension to full Linear Logic

Proof-nets for unit-free Multiplicative Linear Logic

- Formula

$A, B ::= X \mid X^\perp \mid A \otimes B \mid A \wp B$

- Orthogonal

$(X^\perp)^\perp = X \quad (A \otimes B)^\perp = B^\perp \wp A^\perp$
 $(A \wp B)^\perp = B^\perp \otimes A^\perp$

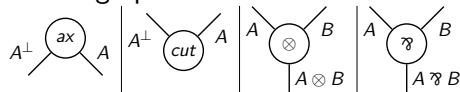
Proof-nets for unit-free Multiplicative Linear Logic

- Formula**

$A, B ::= X \mid X^\perp \mid A \otimes B \mid A \wp B$

- Proof-Structure**

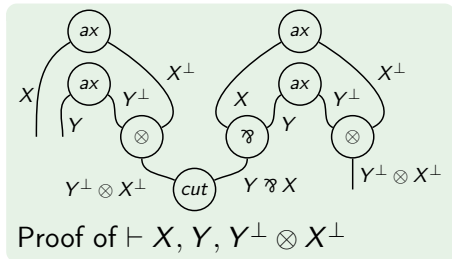
Partial graph built from:



Pending edges = sequent proved

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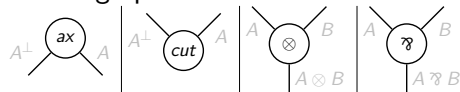
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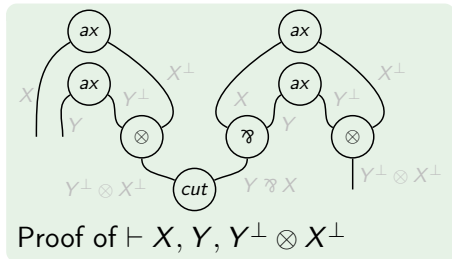


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We do not care about edge labels.

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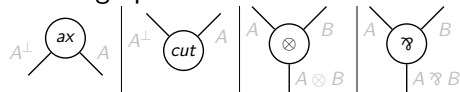
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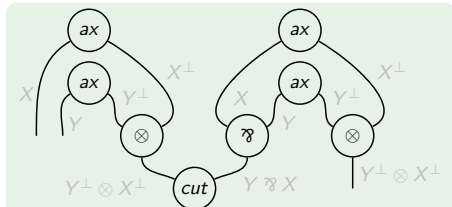
- Proof-Net**

Danos-Regnier correctness criterion:
each cycle contains the two edges
above some $\wp$

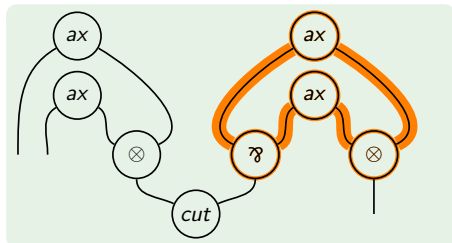
We work with *mix* to simplify, but
everything also holds without it.

- Orthogonal**

$$(X^\perp)^\perp = X \quad (A \otimes B)^\perp = B^\perp \wp A^\perp \\ (A \wp B)^\perp = B^\perp \otimes A^\perp$$

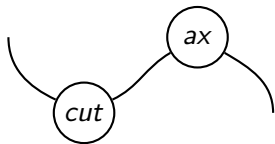


Proof of $\vdash X, Y, Y^\perp \otimes X^\perp$

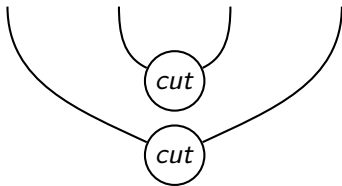
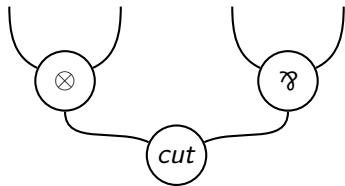


Cut-Reduction

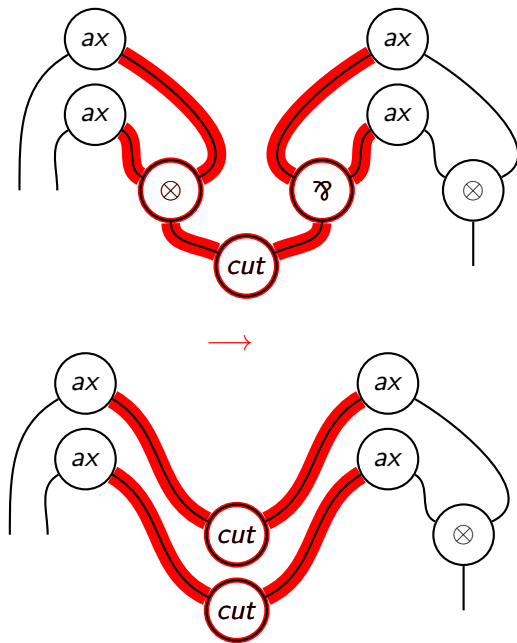
- ax case



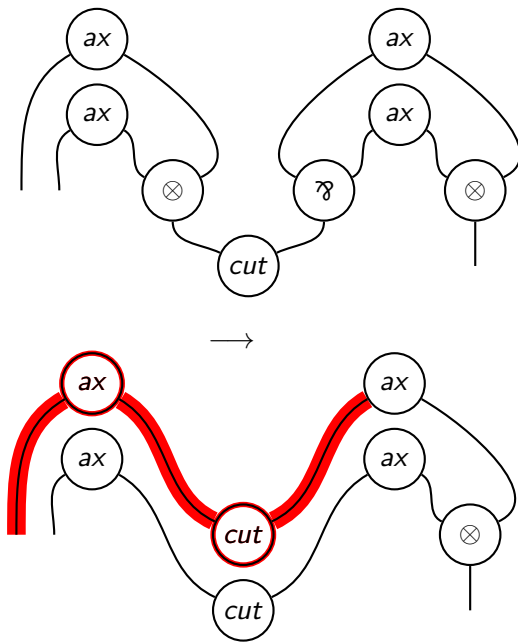
- $\wp - \otimes$ case



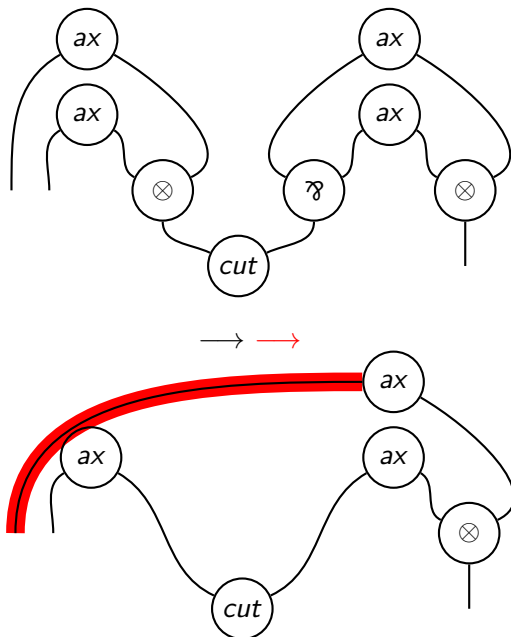
Cut-reduction on an example



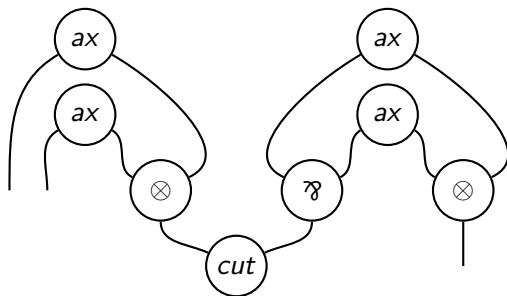
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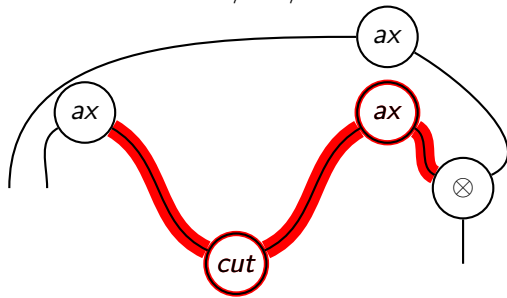
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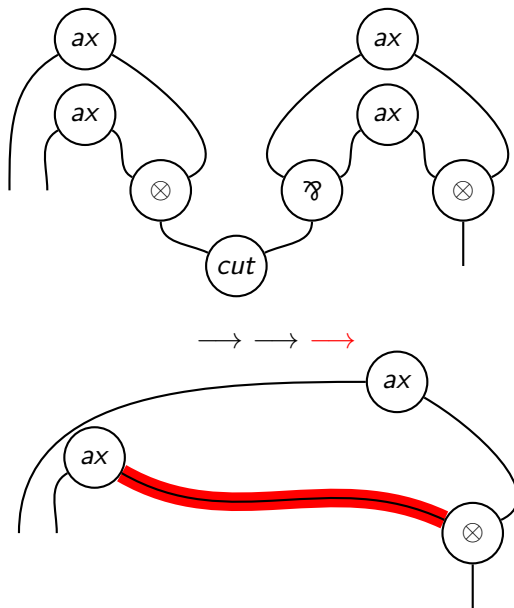
Cut-reduction on an example



$\longrightarrow$



Cut-reduction on an example



On Correctness and Cut-expansion

Has a cut? Can always reduce it.

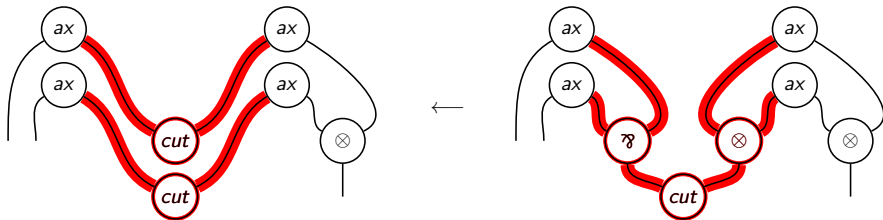
Lemma

If $\mathcal{P}$ is a proof-net and $\mathcal{P} \rightarrow \mathcal{Q}$, then $\mathcal{Q}$ is a proof-net.

Has 2 cuts? Can always expand them.

Fact

*If $\mathcal{P}$ is a proof-net and $\mathcal{P} \leftarrow \mathcal{Q}$, then $\mathcal{Q}$ **may not be** a proof-net.*



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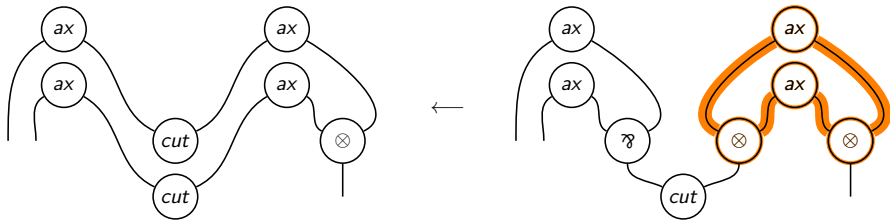
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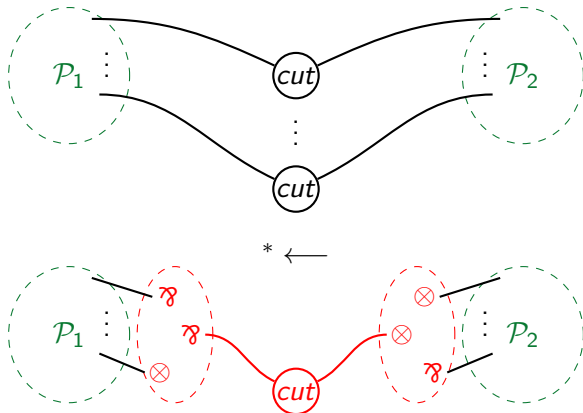


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- ▶ Expanding to a unique *cut* & Applications
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Expanding to a unique cut

Proposition: Expanding to a unique cut

Take a proof-net $\mathcal{P}$ made of two sub-graphs linked by $n \geq 1$ cut-vertices. There is a sequence of $n - 1$ $\wp - \otimes$ cut-expansion steps on these vertices yielding a **proof-net**.



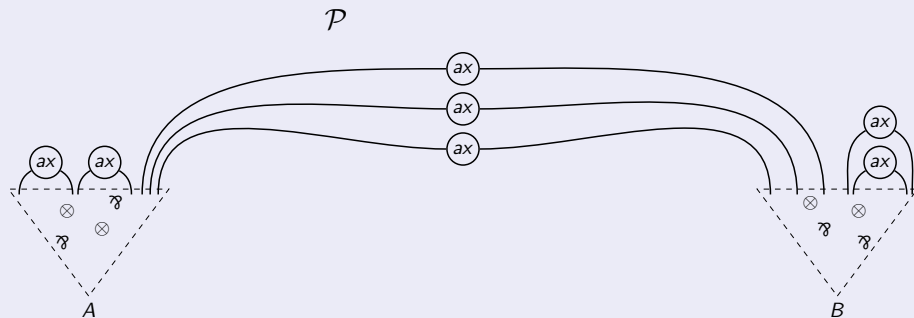
Application: Craig's Interpolation

Proof-relevant Craig's interpolation [Sau25; FOS25]

Set $\mathcal{P}$ a cut-free proof-net of $\vdash A, B$.

There is a formula C whose atoms appear in both A and B and cut-free proof-nets $\mathcal{P}_1$ of $\vdash A, C$ and $\mathcal{P}_2$ of $\vdash C^\perp, B$ such that $\mathcal{P} \ast \leftarrow \mathcal{P}_1$ cut $\mathcal{P}_2$.

New Proof.



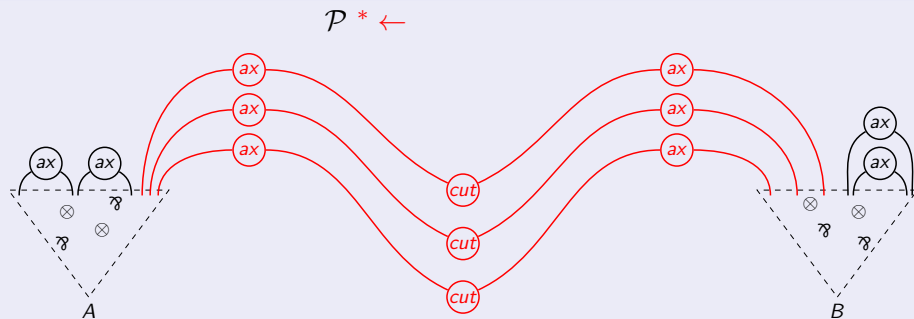
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Application: Craig's Interpolation

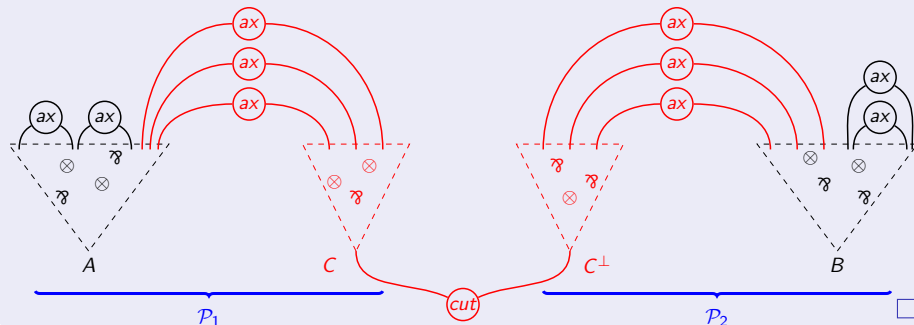
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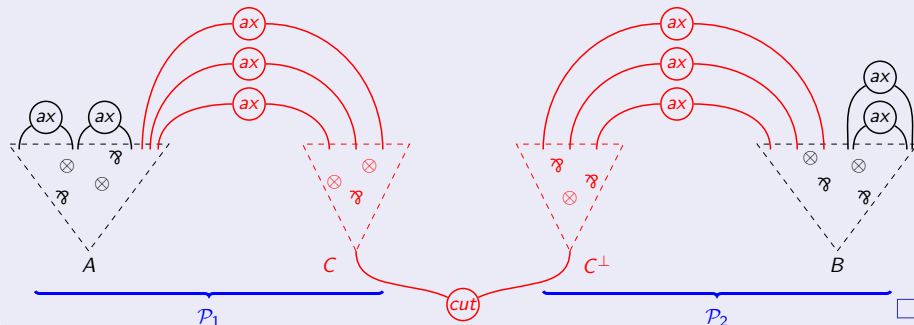
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(Unless $\mathcal{P} = \mathcal{P}_1 \uplus \mathcal{P}_2$ with $\mathcal{P}_1$ of conclusion $\vdash A$ and $\mathcal{P}_2$ of conclusion $\vdash B$.)

New Proof.

$$\mathcal{P}^* \leftarrow \mathcal{P}_1 \text{ cut } \mathcal{P}_2$$



Differences with [FOS25]

Proof-relevant Craig's interpolation [Sau25; FOS25]

Set $\mathcal{P}$ a **connected** cut-free proof-net of $\vdash A, B$.

There is a formula C whose atoms appear in both A and B and cut-free proof-nets $\mathcal{P}_1$ of $\vdash A, C$ and $\mathcal{P}_2$ of $\vdash C^\perp, B$ such that $\mathcal{P} \ast \leftarrow \mathcal{P}_1$ cut $\mathcal{P}_2$.

No mix-rule $\implies$ connected proof-net.

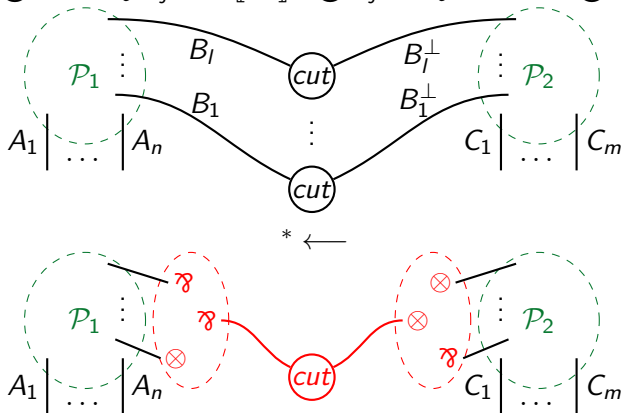
[FOS25] also proves Craig's interpolation in proof-nets but:

- it needs the **units** $\perp$ and 1 , even if $\mathcal{P}$ has none;
- it proceeds by **parsing** (*i.e.* goes top-down) so cannot have **mix**;
meanwhile, we will proceed by **splitting vertices** (*i.e.* bottom-up).

Application: Denotational Semantic

Useful for **compositionality** (e.g. in [EFP24]): given *interpretations* $\llbracket \mathcal{P}_1 \rrbracket$ and $\llbracket \mathcal{P}_2 \rrbracket$, how to get the *interpretation* of the full net?

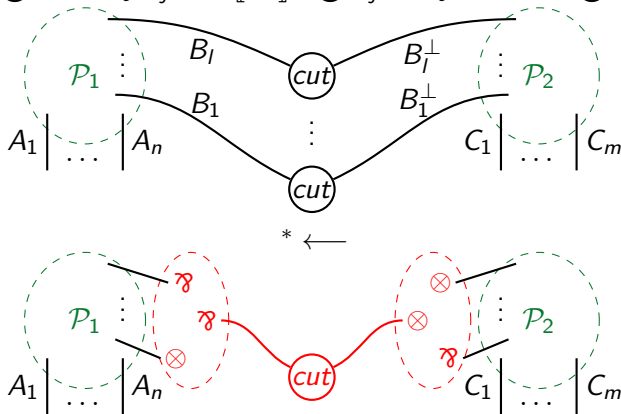
↪ Easy when **only one cut**, otherwise *typing* problem: given $\llbracket \mathcal{P}_1 \rrbracket : \bigotimes A_i \rightarrow \wp B_j$ and $\llbracket \mathcal{P}_2 \rrbracket : \bigotimes B_j \rightarrow \wp C_k$, want $\bigotimes A_i \rightarrow \wp C_k$



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[EFP24]'s solution with the *same result* as here, but using **parsing** again!

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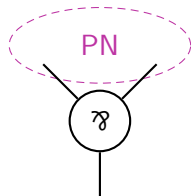
Reasoning

Proposition: Expanding to a unique *cut*

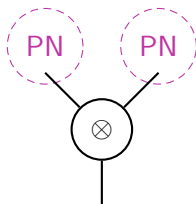
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- Proof by induction on $\#|\text{vertices}|$
- Main consideration: when adding a $\otimes$, we ensure it creates no “bad cycle”
- Case study using a proof-net with a *cut* always has a **splitting vertex**:

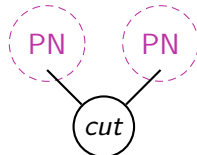
• $\wp$ -vertex



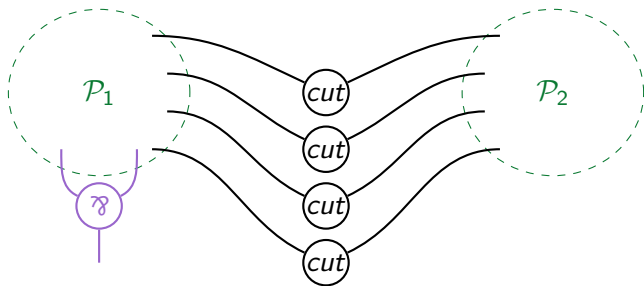
• $\otimes$ -vertex



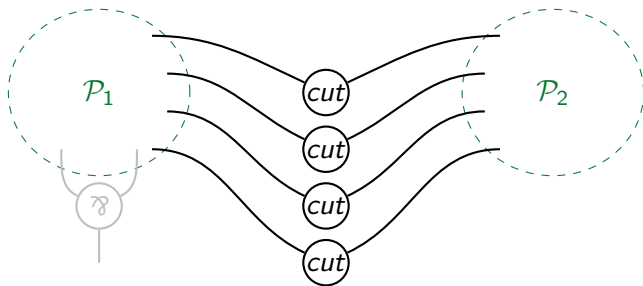
• *cut*-vertex



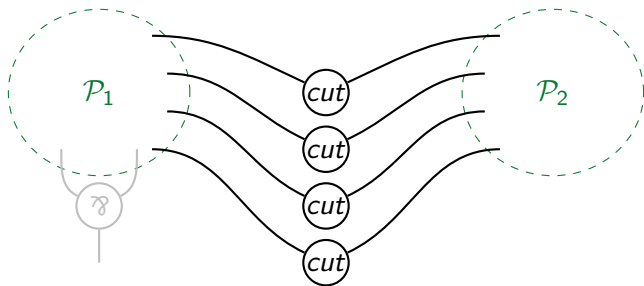
Splitting $\mathcal{P}$



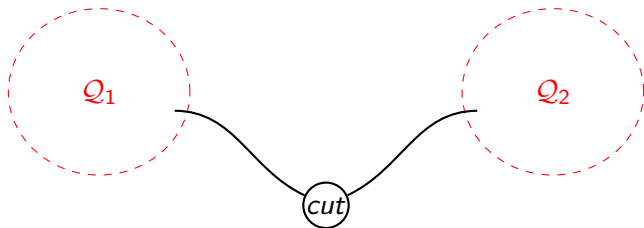
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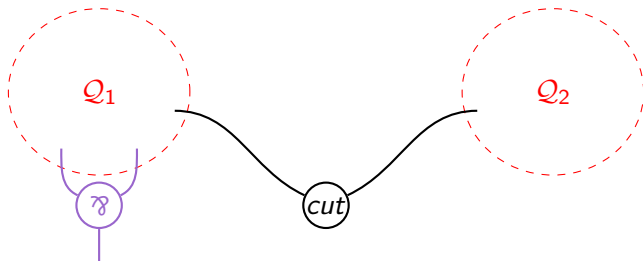
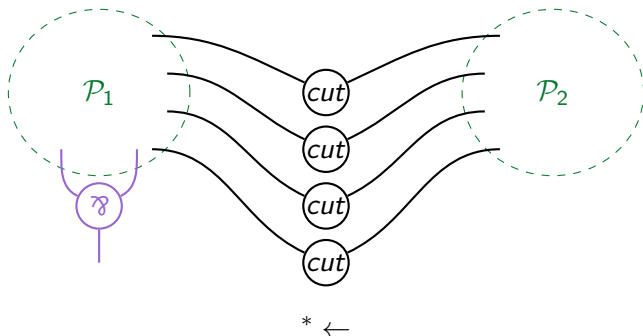
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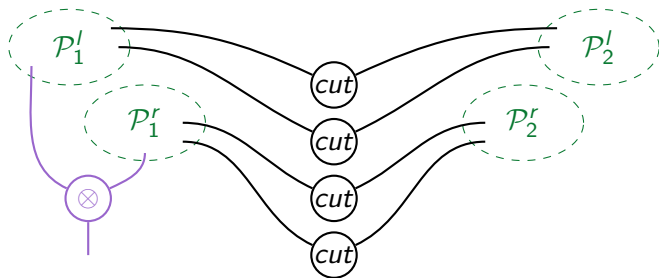
* $\leftarrow$



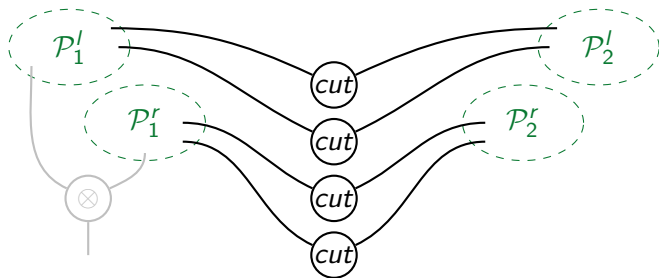
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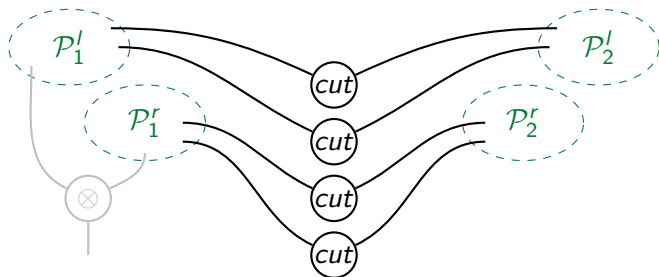
Splitting $\otimes$



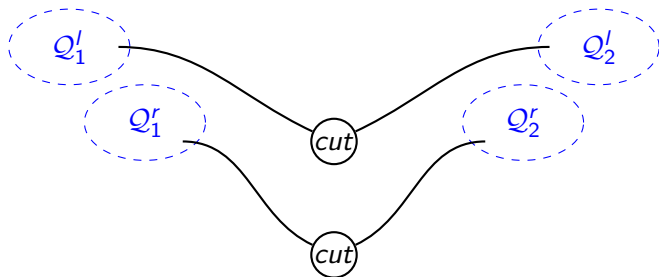
Splitting $\otimes$



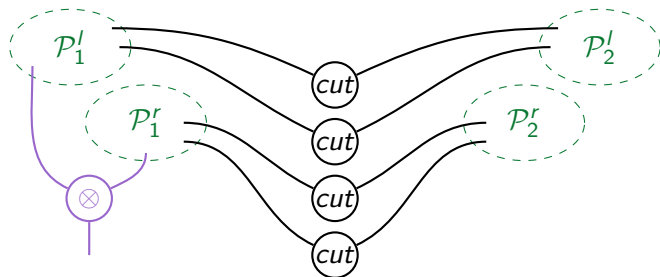
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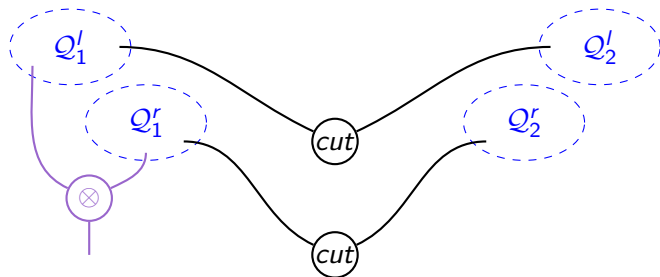
$* \leftarrow * \leftarrow$



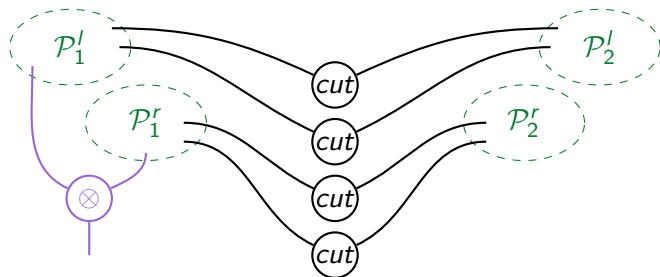
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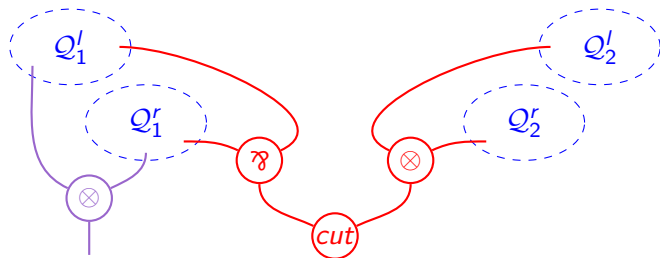
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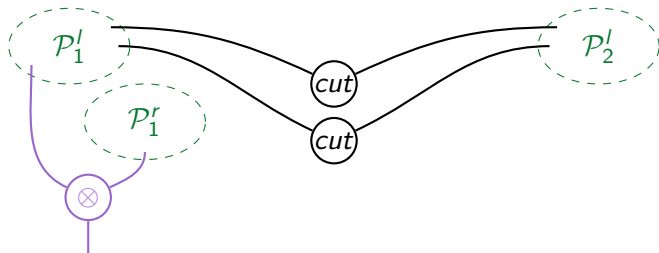
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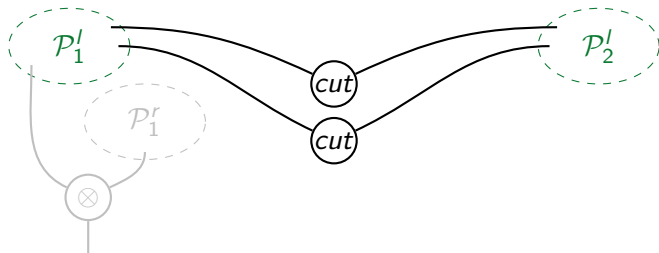
* $\leftarrow$ * $\leftarrow$ $\leftarrow$



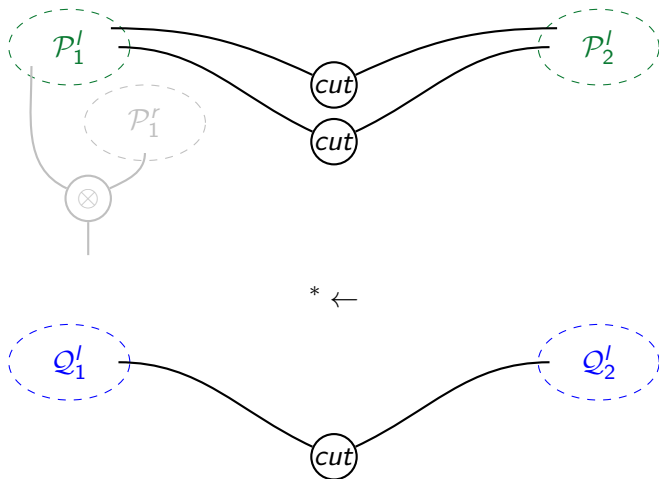
Splitting $\otimes$ *bis*



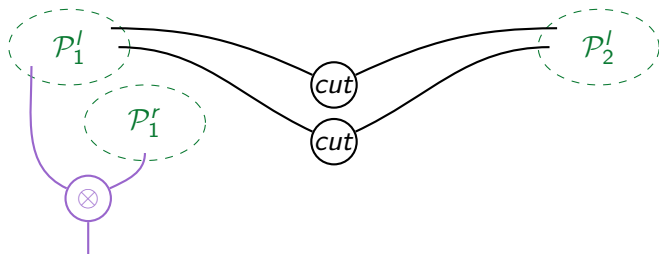
Splitting $\otimes$ *bis*



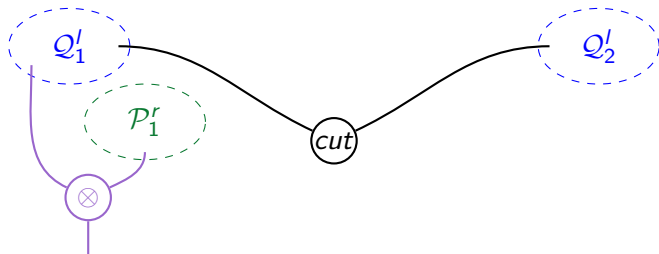
Splitting $\otimes$ *bis*



Splitting $\otimes$ *bis*

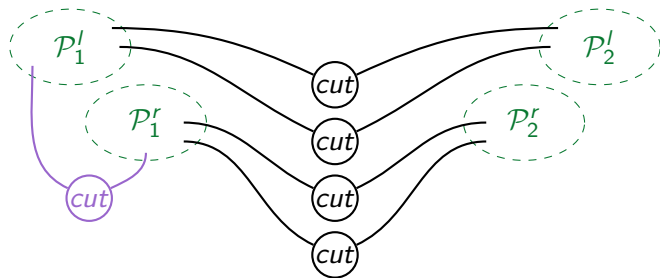


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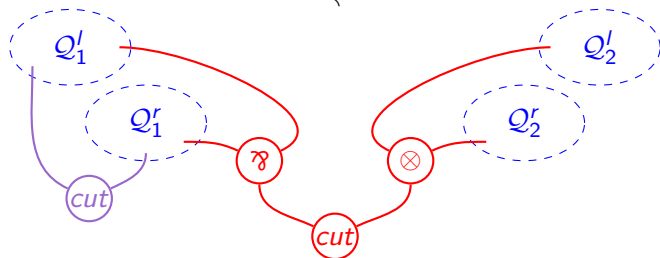


Splitting *cut*

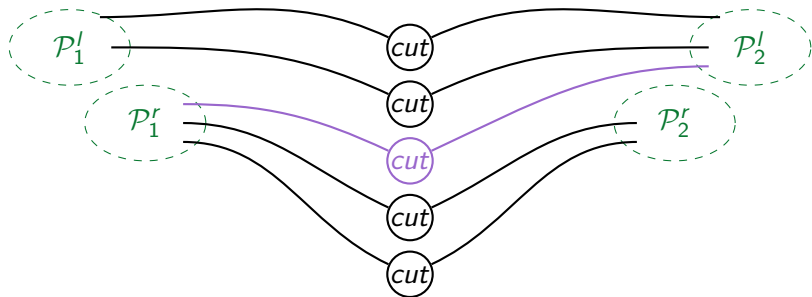
If this *cut* is not between the two sub-graphs: same as the $\otimes$ case.



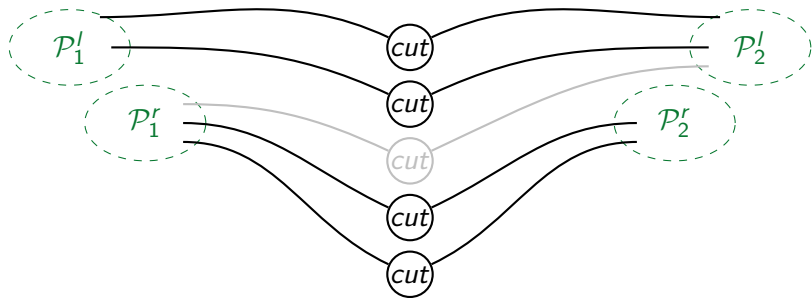
* ←



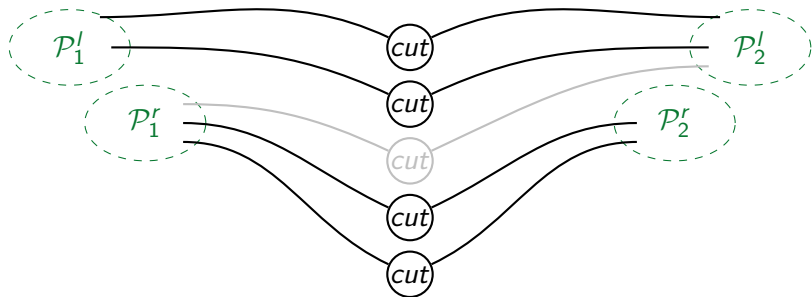
Splitting *cut*



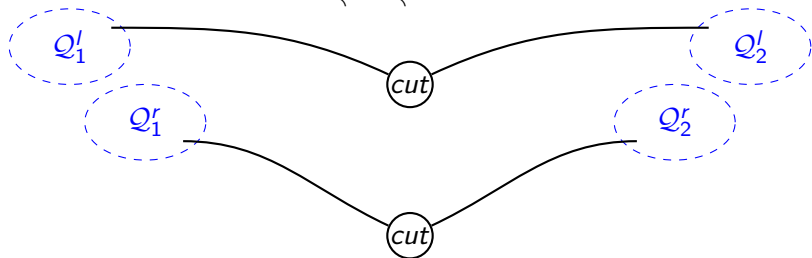
Splitting *cut*



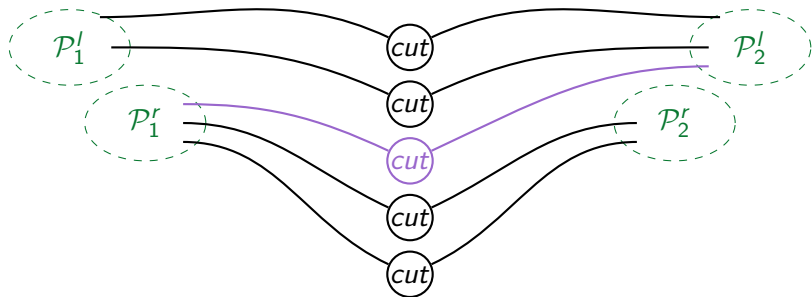
Splitting *cut*



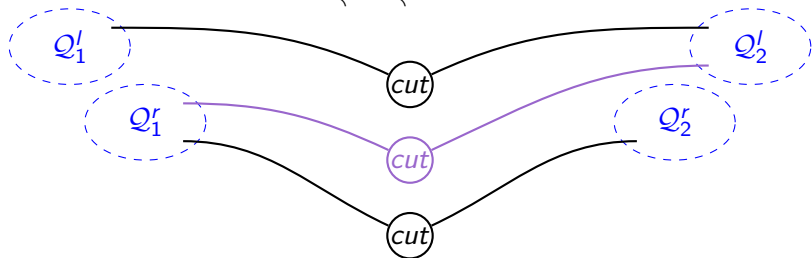
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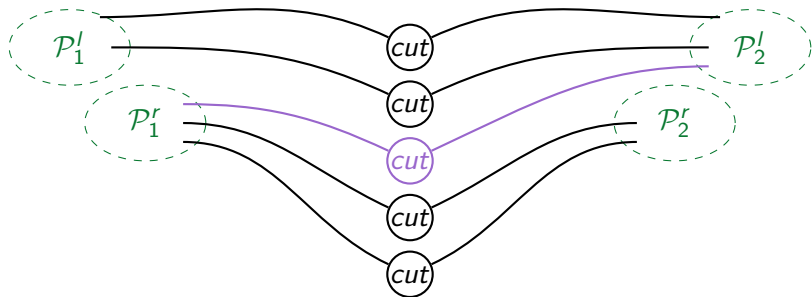
Splitting *cut*



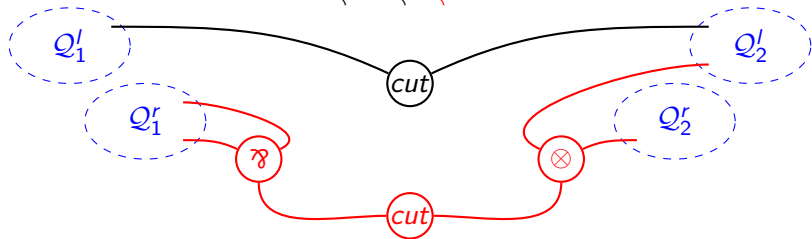
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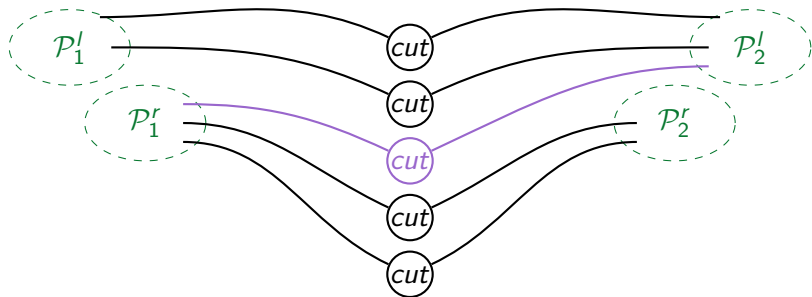
Splitting *cut*



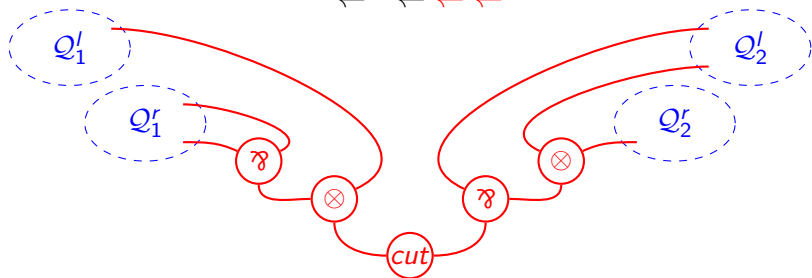
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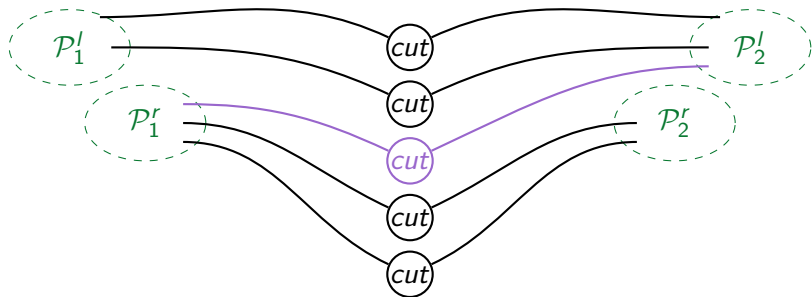
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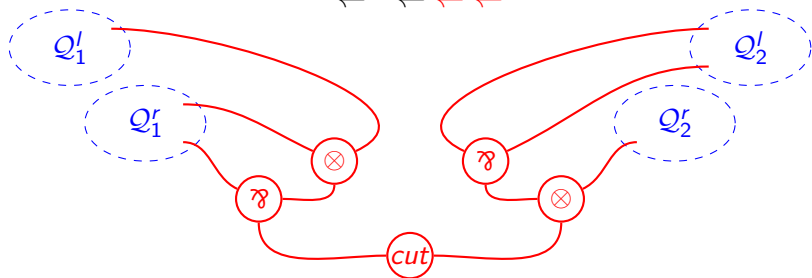
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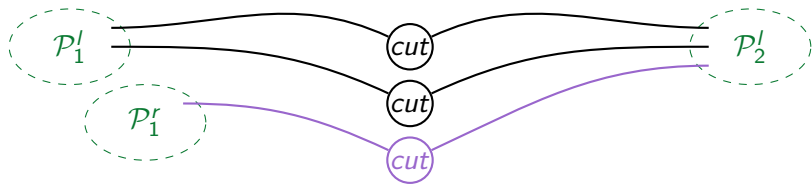
Splitting *cut*



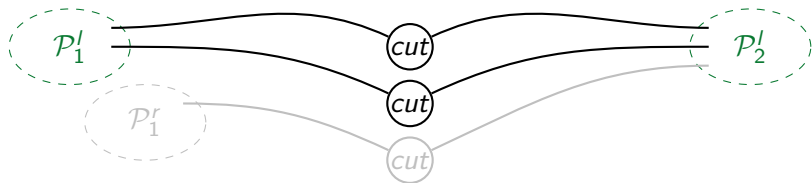
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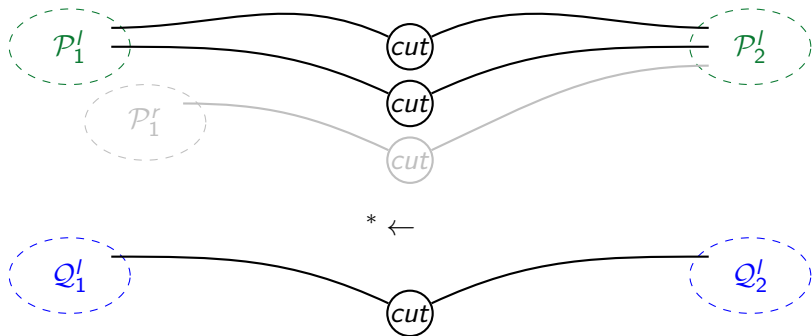
Splitting *cut bis*



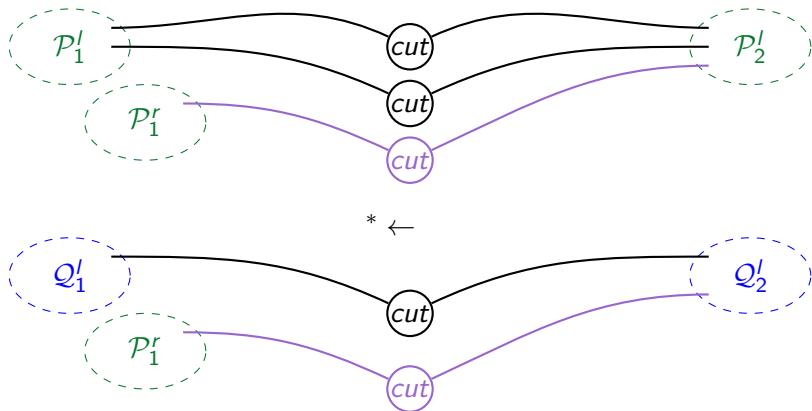
Splitting *cut bis*



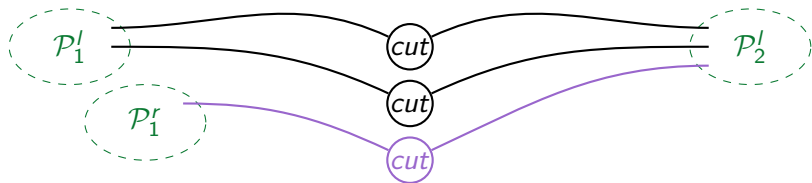
Splitting *cut bis*



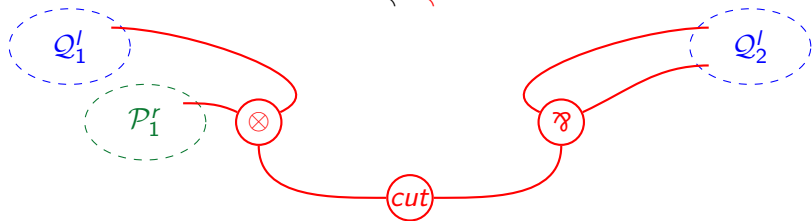
Splitting *cut* *bis*



Splitting *cut bis*



* $\leftarrow$ $\leftarrow$



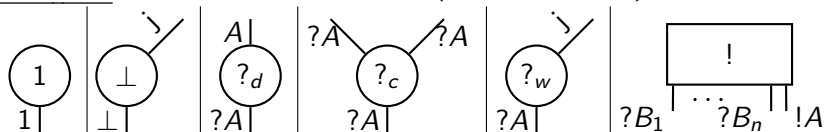
Plan

- ▶ Proof-Nets & Cut-Reduction
- ▶ Expanding to a unique *cut* & Applications
 - Craig's Interpolation
 - Denotational Semantic
- ▶ Proof of the Expansion to a unique *cut*
- ▶ Extension to full Linear Logic

In Proof-Nets for MLL with units & MELL

Same as for unit-free MLL!

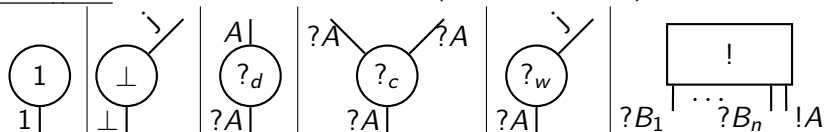
- Proof-nets with jumps for $\perp$ and $?_w$ (or the mix-rules) and boxes for $!$:



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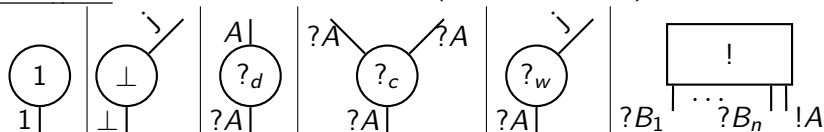
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Same as before: $\perp$, $?_d$, $?_c$, $?_w$ treated as a $\wp$, 1 and $!$ irrelevant (like ax)

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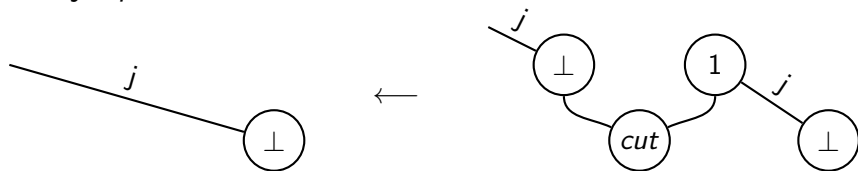


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- Craig's interpolation just needs additional “pre-processing” steps:

- ▶ for a *jump* between A and B :



- ▶ for a $!$ -box using both A and B : a little bit more complex, uses an *induction* (on the net associated to the promotion), then move this *cut* out of the box (by a $?_d - !$ step followed by a $! - !$ step)

In Sequent Calculus for full LL: what are cut-nets?

No proof-net for full LL $\longrightarrow$ go to sequent calculus

Definition: Derivation corresponding to a cut-net

A proof derivation π of $\vdash \Gamma$ is a **cut-derivation** if, denoting (A_i) the cut-formulas of π , there is a bipartition $\{B_1, B_2\}$ of $\Gamma \cup (\bigcup_i A_i)$ such that:

- all ax -rules are applied either only on sub-formulas of B_1 or only on sub-formulas of B_2 ; and similarly for $!-$ and $\top$ -rules;
 $\leadsto$ no ax , $!-$ or $\top$ using both sides
- given a $\perp$ -rule associated to B_i , the rule just above is not a *cut*-rule and has for principal formula a sub-formula of B_i or is an $ax/!/ \top$ -rule between sub-formulas of B_i ; and similarly for a $?_w$ -rule;
 $\leadsto$ no $\perp$ - or $?_w$ -jump going from one side to the other
- given an $\exists$ -rule associated to B_i , its witness has no atoms introduced by a $\forall$ -rule associated to B_{1-i} .
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Deduce Craig's interpolation as before: introduce a cut when one of the hypothesis of a *cut-derivation* is not respected.

In Sequent Calculus for full LL: proof

Proposition: Expanding to a unique cut

Take a cut-derivation π . There is a sequence of cut-expansion steps yielding a proof with only one cut-rule, at its root.

Similar proof as previously: *case study on a last rule*

!! & case: $\vdash A \& B, \Gamma_1, \Gamma_2$ where $\Gamma_2 = \emptyset$ and only one branch has a cut:

$$\frac{\frac{\vdash A, \Gamma_1, C_1 \quad \vdash C_1^\perp}{\vdash A, \Gamma_1} \text{ (cut)}}{\vdash A \& B, \Gamma_1} \text{ (&)}$$

How to move the cut at the root?

In Sequent Calculus for full LL: proof

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How to move the *cut* at the root?

- Introduce a *cut* on $\perp - 1$ in the right branch: *no more stable by fragment*; or
- Eliminate the *cut* in the left branch: *no more only cut-expansion*.

~> The additive connectives need the multiplicative units to work well, but the multiplicative connectives do not! (Quantifiers also need them.)

Conclusion

- Simple visual proof of expansion to a unique *cut*, with and without mix
- Yields a proof of Craig's interpolation for unit-free MLL, MLL and MELL
- Useful for denotational semantic when typing an interface
- Adaptable in sequent calculus for full linear logic, but definitions less visual and some problems with $\&$ and quantifiers

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Thank you!

References I

- [EFP24] Thomas Ehrhard, Claudia Faggian, and Michele Pagani. *Bayesian Networks and Proof-Nets: a proof-theoretical account of Bayesian Inference*. 2024. DOI: [10.48550/ARXIV.2412.20540](https://doi.org/10.48550/ARXIV.2412.20540). URL: <https://arxiv.org/abs/2305.18945>.
- [FOS25] Guido Fiorillo, Daniel Osorio Valencia, and Alexis Saurin. “On Correctness, Sequentialization and Interpolation”. In: *Ninth International Workshop on Trends in Linear Logic and Applications 2025*. Ed. by Lionel Vaux Auclair. 2025. URL: <https://lipn.univ-paris13.fr/TLLA/2025/abstracts/14-fiorillo-osoriovalencia-saurin.pdf>.

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